

3.1 – Pairwise Test Design for schedule_appointment()

Parameter	Values (levels)
appt_type	new_patient, follow_up, urgent_care, procedure, telehealth (5 values)
duration_minutes	15, 30, 45, 60 (4 values)
day_of_week	Monday (valid), Wednesday (valid), Friday (valid), Saturday (invalid) (4 values)
hour_of_day	8 (open boundary), 10 (midday), 16 (close boundary), 17 (just closed) (4 values)

(a) Full Factorial Calculation

Appointment Type = 5 values

Duration = 4 values

Day of Week = 4 values

Hour of Day = 4 values

Full factorial = $5 \times 4 \times 4 \times 4 = 320$ tests.

Pairwise reduces this to approximately 20 tests

(b) Pairwise Coverage Table (20 Test Rows)

TC	Appointment Type	Duration	Day	Hour	Pairs Covered
1	new_patient	15	Monday	8	(new_patient,15), (new_patient,Monday), (new_patient,8), (15,Monday), (15,8), (Monday,8)
2	new_patient	30	Wednesday	10	(new_patient,30), (new_patient,Wednesday), (new_patient,10), (30,Wednesday), (30,10), (Wednesday,10)
3	new_patient	45	Friday	16	(new_patient,45), (new_patient,Friday), (new_patient,16), (45,Friday), (45,16), (Friday,16)
4	new_patient	60	Saturday	17	(new_patient,60), (new_patient,Saturday), (new_patient,17), (60,Saturday), (60,17), (Saturday,17)
5	follow_up	15	Wednesday	16	(follow_up,15), (follow_up,Wednesday), (follow_up,16), (15,Wednesday), (15,16), (Wednesday,16)
6	follow_up	30	Friday	17	(follow_up,30), (follow_up,Friday), (follow_up,17), (30,Friday), (30,17), (Friday,17)
7	follow_up	45	Saturday	8	(follow_up,45), (follow_up,Saturday), (follow_up,8), (45,Saturday), (45,8), (Saturday,8)
8	follow_up	60	Monday	10	(follow_up,60), (follow_up,Monday), (follow_up,10), (60,Monday), (60,10), (Monday,10)
9	urgent_care	15	Friday	10	(urgent_care,15), (urgent_care,Friday), (urgent_care,10), (15,Friday), (15,10), (Friday,10)
10	urgent_care	30	Saturday	16	(urgent_care,30), (urgent_care,Saturday), (urgent_care,16),

					(30,Saturday), (30,16), (Saturday,16)
11	urgent_care	45	Monday	17	(urgent_care,45), (urgent_care,Monday), (urgent_care,17), (45,Monday), (45,17), (Monday,17)
12	urgent_care	60	Wednesday	8	(urgent_care,60), (urgent_care,Wednesday), (urgent_care,8), (60,Wednesday), (60,8), (Wednesday,8)
13	procedure	15	Saturday	10	(procedure,15), (procedure,Saturday), (procedure,10), (15,Saturday), (15,10), (Saturday,10)
14	procedure	30	Monday	16	(procedure,30), (procedure,Monday), (procedure,16), (30,Monday), (30,16), (Monday,16)
15	procedure	45	Wednesday	17	(procedure,45), (procedure,Wednesday), (procedure,17), (45,Wednesday), (45,17), (Wednesday,17)
16	procedure	60	Friday	8	(procedure,60), (procedure,Friday), (procedure,8), (60,Friday), (60,8), (Friday,8)
17	telehealth	15	Monday	17	(telehealth,15), (telehealth,Monday), (telehealth,17), (15,Monday), (15,17), (Monday,17)
18	telehealth	30	Wednesday	8	(telehealth,30), (telehealth,Wednesday), (telehealth,8), (30,Wednesday), (30,8), (Wednesday,8)
19	telehealth	45	Friday	10	(telehealth,45), (telehealth,Friday), (telehealth,10), (45,Friday), (45,10), (Friday,10)
20	telehealth	60	Saturday	16	(telehealth,60), (telehealth,Saturday), (telehealth,16), (60,Saturday), (60,16), (Saturday,16)

(c) Implementation

The pairwise test cases were implemented in `tests/test_hw5_combinatorial.py` using `pytest.mark.parametrize`. Each row in the table corresponds to one parameterized test execution (20 total tests).